

Course syllabus for

Academic year 2022/2023 >

FUF050 - Subatomic physics

Subatomär fysik

Course syllabus adopted 2021-02-26 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

6,0 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Engineering Physics

Department: 16 - PHYSICS

Teaching language: Swedish

Application code: 57113

Open for exchange students: No

Block schedule: [C+](#)

Maximum participants: 130

Only students with the course round in the programme overview

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0106 Examination 6,0 c Grading: TH					6,0 c		Tue 30/05-2023 am J, Fri 07/10-2022 pm J, Thu 24/08-2023 pm J

In programs

[TKTFY ENGINEERING PHYSICS, Year 3 \(compulsory\)](#)

Examiner:

[Andreas Ekström](#)

[Go to course homepage](#)

Theme:

Environment 1,5 hec

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

Quantum physics F3

Aim

Subatomic physics covers the area of physics that describes the composition of the atomic nucleus as well as its properties and interaction with matter. The subject, earlier known as nuclear physics, has been enlarged to contain the interface between nuclear and particle physics and also nuclear astrophysics. World-wide intensive research activities are going on both theoretically and experimentally.

Based on the knowledge of quantum physics acquired in earlier courses the aim of the course is to give an introduction to the subatomic part of modern physics.

Learning outcomes (after completion of the course the student should be able to)

The goal is to develop specific knowledge about the constituents of and structure of the atomic nucleus and to provide understanding of subatomic processes. The student will further acquire an introductory knowledge of phenomenological models describing the nucleus. The basics of elementary particle physics will be introduced and also intermediate energy phenomena at the border between nuclear and particle physics. Nuclear synthesis and energy production in the sun are also included. Description of experimental techniques will make the student

acquainted with scientific methodology.

Understanding radioactive decay processes and the passage of radiation through matter provides important insights about sustainability and the environment.

Content

The most important sections of the course are radius and mass, nuclear stability, radioactivity, alpha- and beta-decay, gamma emission, nuclear reactions, fission, fusion, nuclear models, elementary particles, cosmology, nuclear astrophysics and experimental techniques.

Organisation

Teaching is mediated in lectures, problem solving sessions and laboratory work. The laboratory exercises normally include beta spectroscopy, coincidence measurements and neutron activation of silver.

Literature

Communicated minimum two weeks ahead of course

Examination including compulsory elements

Written examination, three laboratory experiments, and voluntary assignments in the form of written reports. The reports can give bonus points.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.